electrons. However, the spurs are quite close to each other. For the case of α-particles, the original distance between spurs is of the order of fractions of a nanometer. Immediately at their formation they comprise a continuous cylindrical column. The column consists of a dense core surrounded by a more diffuse shell of tracks of high energy δ-electrons.

8.3. RADIATION DOSE AND RADIATION YIELD

The oldest radiation unit still in use is the *roentgen* (R). It applies only to photons and is defined as the *exposure*. The exposure is the energy flux of the unperturbed photon radiation when hitting matter. 1 R is the exposure that in air produces ion pairs with total charge per unit mass of 2.58×10^{-4} C/kg. This corresponds to 1.61×10^{15} ion pairs per kg air or 8.8×10^{-3} J/kg absorbed energy at an average ion pair formation energy of 34 eV. The roentgen has been in use for more than 60 years and is still not uncommon in the medical profession in some countries.

The *absorbed dose* (D) is the amount of radiation energy absorbed per unit mass:

$$D = \mathrm{d}E_{\mathrm{abs}}/\mathrm{d}m \tag{8.9}$$

According to Eqn (8.1), in the absense of nuclear transformations, $E_{\mathrm{abs}} = E_{\mathrm{in}} - E_{\mathrm{out}}$. The SI unit is the *Gray* (Gy)

$$1\ \mathrm{Gy} = 1\ \mathrm{J/kg} \tag{8.10}$$

An old, more or less obsolete, unit is the *rad* (for radiation absorbed dose)

$$1\ \mathrm{Gy} = 100\ \mathrm{rad} \tag{8.11}$$

The rad was proposed in 1918 as "that quantity of X rays which when absorbed will cause the destruction of the malignant mammalian cells in question..." Formally it was defined in 1953 as the dose causing 100 ergs of energy to be absorbed by one gram of matter. An erg is the unit of energy and mechanical work in the old centimetre-gram-second (CGS) system of units. The name comes from the greek word for work, ergon. One erg is the amount of work done by a force of one dyne exerted for a distance of one centimeter. The dyne also originates from a greek word, namely dynalis meaning force or power. The dyne is defined as "the force required to accelerate a mass of one gram at a rate of one centimeter per second squared" Although these units may still be found in the literature their use is generally discouraged in education for other pursposes than for relating old measures to units in the SI system. In some countries, however, they are still used as an industrial standard.

The *dose rate* is the absorbed dose per unit time. The SI unit is Gray per second (Gy/s).

The *specific γ-ray dose rate,* $\dot{D}$, is a practical measure for estimation of the radiation hazard to people from γ-emitting radionuclides:

$$\dot{D} = A\, r^{-2} \Sigma\, n_i\, k_i\, B_i\, \mathrm{e}^{-\mu_i x} \tag{8.12}$$

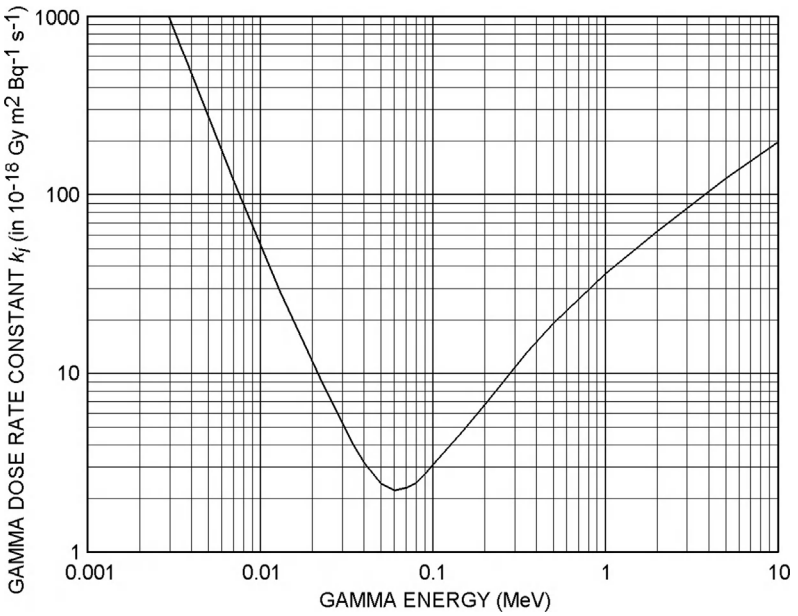


Figure 8.2 Dose rate constant for a monoenergetic point isotropic γ-source as a function of γ-energy.

where $A\ r^{-2}\ \Sigma\ n_i\ k_i = \dot{D}_0$ is the relative dose rate (Gy/s) without any radiation shielding, and the same for a radiation shield with build-up factor B_i and attenuation factor $e^{-\mu_i x}$ (see eqns. 7.8 and 7.27—7.28); n_i is the fraction of all decays (given by the decay scheme of the source nuclide) yielding a γ-ray of energy E_i corresponding to the source constant k_i according to Figure 8.2, and A is the source strength (Bq). E.g. ^{24}Ne decays to 8% with two γ's (0.878 and 0.472 MeV) and to 92% with one γ (0.472 MeV). Thus the sum in Eqn (8.12) contains two terms, one for 0.878 MeV γ ($n_1 = 0.08$) and one for 0.472 MeV γ ($n_2 = 1$). The k_i-values are taken from Figure 8.2. The relaxation length $\mu_i\ x$ is computed for each energy E_i and the corresponding attenuation factors of the radiation shield $e^{-\mu_i x}$(x m thick) are calculated using e.g. Figure 7.17. Build-up factors B_i can be estimated from Figure 7.20. It should be noted that Eqn (8.12) does not account for scattering around the shield ("sky-shine"), which has to be estimated separately; see text books on radiation shielding.

Closely related to the absorbed dose is the *kerma* (K), which is the kinetic energy released per unit mass by uncharged particles and electromagnetic radiation (n and γ):

$$K = \mathrm{d}E_{\mathrm{tr}}/\mathrm{d}m = \varphi\ E\ \mu_{\mathrm{tr}}\ \rho^{-1} \tag{8.13}$$

where E is the radiation energy, φ is the particle fluence (particles/m^2) and μ_{tr}/ρ is the mass energy transfer coefficient. The SI unit of the kerma is J/kg.

In the following we refer to all absorbed radiation energy as the radiation dose independent of whether the incident radiation is charged or uncharged particles or photons.

Radiation chemical yield is described in terms of *G-values*. Originally $G(x)$ was defined as the number of molecules of x transformed per 100 eV absorbed energy (a practical notation as most reactions have G-values of <10). In the SI system the symbol $G(x)$ is the same but the unit is mol/J. The conversion factor between the two units is 1 mol/J $= 9.649 \times 10^6$ molecules per 100 eV absorbed. Today the symbol $g(x)$ usually refer to the number per 100 eV and $G(X)$ to mol/J.

8.4. METALS

Metals consist of a solid lattice of atoms whose valence electrons cannot be considered to belong to any particular atom, but rather to a partially filled energy band (the conduction band) established by the total lattice network. Interaction of radiation with the metal can cause *excitation* of bound electrons in the atoms to the conduction band. This, however, due to the excellent conducting properties of metals do not lead to prevailing electron deficiencies resulting in e.g. colour changes.

While irradiation by γ-rays and electrons has little influence on metallic properties, heavy particles cause serious damage through their collision with atoms in the metal lattice network. This results in *displacements* of the atoms from their lattice positions. The number of displacements (n_{disp}) depends on the amount of energy transferred in the collision event (E_{tr}) to the recoiling (target) atom, and the energy required for moving this atom from its lattice position. This so-called *displacement energy* averaged over all directions in the crystal lattice (E_{disp}) is 10−30 eV for most metallic materials. According to the Kinchin-Pease rule,

$$n_{\text{disp}} \leq \tfrac{1}{2} E_{\text{tr}} / E_{\text{disp}} \tag{8.14}$$

The maximum energy transferred can be calculated assuming purely elastic collisions between hard spheres (§10.1). Thus for a 1.5 MeV fission neutron, E_{tr}(max) is 425 keV in C, 104 keV in Fe, and 25 keV in U. With $E_{\text{disp}} \approx 25$ eV, up to 8500, 2080 and 500 displacements, respectively, occur in these metals due to the absorption of a fission neutron (neglecting nuclear reactions). In practice the numbers are somewhat smaller, especially at the higher E_{tr} (where it may be about one third of the calculated value). The reason for this is the recombination of defects at the end of the collision cascade. Atomic displacements cause many changes in the properties of metals. Usually electrical resistance, volume, hardness and tensile strength increase, while density and ductility decrease.

The microcrystalline properties of metals are particularly influenced by irradiation. Although low-alloy steel in modern reactor tanks are rather radiation resistant (provided